

TRADE ASSOCIATION NETWORKS AND GLOBAL NTM TRADE POLICY FORMATION

Community Data Lab Project Report

World Bank

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Summary

This report outlines the progress made in constructing a unified dataset to analyze non-tariff measure (NTM) proliferation at the industry level. The project integrates product-level trade policy data with industry-level organizational and economic data to create a country–year–industry (ISIC Rev. 3, 4-digit) panel suitable for empirical analysis.

Because the NTM data are recorded at the Harmonized System (HS) product level while trade association and industry statistics use ISIC classification, significant data harmonization is required. The data construction process involves collapsing NTM observations to the country–year–HS level, mapping HS codes to ISIC Rev. 3 using official concordance tables, and aggregating the data to the country–year–industry level. In parallel, INDSTAT data are cleaned to remove 3-digit industries and reshaped so that industry characteristics appear as columns rather than rows. The final deliverable is a merged panel dataset that combines NTM intensity measures, trade association density, and industry-level economic indicators.

At this stage, the concordance mapping structure has been implemented and the data pipeline for collapsing, reshaping, and merging has been established in Python. Remaining tasks include completing mapping across all HS nomenclature versions, validating merge consistency, generating descriptive statistics, and preparing regression specifications for subsequent analysis.

Introduction and Research Question

This project investigates the rapid global rise in non-tariff measures (NTMs) since 2000 and examines whether the concentration of NTMs across industries is associated with the density of trade associations within those industries. The central objective is to evaluate whether industries with dense trade association networks experience earlier or more intense NTM proliferation.

The main research question guiding this study is:

“Do industries with greater trade association density experience higher levels of non-tariff trade protection across countries and over time?”

This question is motivated by political economy theories of trade policy formation. As tariffs decline due to WTO commitments and international agreements, governments may substitute toward non-tariff measures as an alternative protection tool. Industries with stronger lobbying capacity, potentially proxied by trade association density, may be more effective in influencing trade policy outcomes. The project aims to construct a country–year–industry panel dataset that links NTM intensity to industry-level trade association density and other industrial characteristics.

Data Description

This project combines multiple datasets at different classification levels, requiring significant data construction before analysis.

NTM Data (WITS)

The non-tariff measure data come from the World Integrated Trade Solution (WITS), a joint initiative of the World Bank and UNCTAD. The dataset records NTMs applied by countries at the product level using Harmonized System (HS) product codes. Each observation includes:

- Reporting country (country implementing the NTM)
- Partner country (if specified)
- HS product code
- HS nomenclature version (H2, H3, H4, H5)
- NTM type
- Start year

As noted in the project documentation, HS nomenclature changes over time, and the data include multiple HS revisions. Therefore, concordance tables must be used to convert HS codes into ISIC Revision 3 industry codes.

Trade Association Data

Trade association data are drawn from the World Guide to Trade Associations, which has been digitized and structured into country–industry format. These data provide information on the number and density of trade associations within industries across countries. The key explanatory variable is trade association density at the country–industry level.

Industry-Level Controls (INDSTAT)

Industry-level economic characteristics obtained from the UNIDO database. These include:

- Employment
- Number of establishments
- Output
- Value added
- Wages and salaries

These variables serve as controls in regression analysis to account for industry size and economic importance.

Data Construction Strategy

A core challenge of this project is that the NTM data are recorded at the HS product level, while trade association and INDSTAT data are recorded using ISIC industry codes. A central component of this project is constructing a unified country–year–industry panel dataset that combines product-level NTM data with industry-level characteristics. Because the datasets

originate in different classification systems (HS for NTMs and ISIC for industry data), careful restructuring and aggregation are required before merging.

Collapsing the NTM Data Prior to Crosswalk

The raw NTM dataset is recorded at the product (HS code) level and includes multiple observations per country–year–product combination. The first step is to collapse the NTM data before applying the HS–ISIC concordance crosswalk.

Specifically, the NTM data are first grouped to the level of: Country, Year, and HS product code. Within each country–year–HS group, NTMs are aggregated (e.g., counting the number of measures affecting that HS code). This produces an intermediate dataset structured as:

Country | Year | HS Code | NTM Intensity

Collapsing before the crosswalk reduces duplication and prevents inflated counts that may arise when one HS code maps to multiple ISIC industries. It also significantly improves computational efficiency when working with large NTM datasets.

Mapping HS to ISIC Rev. 3

After collapsing, the dataset is merged with official WITS concordance tables that map HS product codes to ISIC Revision 3 industry codes. Because NTM data span multiple HS nomenclature versions (H2, H3, H4, H5), the following concordances are used:

- H2 → ISIC Rev. 3
- H3 → ISIC Rev. 3
- H4 → ISIC Rev. 3
- H5 → H4 → ISIC Rev. 3

After merging with the concordance tables, ISIC codes are standardized to the 4-digit level (ISIC4) to ensure compatibility with trade association and INDSTAT data. The NTM data are then re-collapsed to:

Country | Year | ISIC4 | NTM Intensity

Cleaning and Reshaping INDSTAT

The INDSTAT dataset is originally structured at the industry level using ISIC codes but may contain mixed levels of aggregation (e.g., 3-digit and 4-digit industries). To ensure consistency:

- 3-digit ISIC observations are removed, retaining only ISIC Rev. 3 4-digit industries.
- The dataset is reshaped (if necessary) so that industry characteristics appear as columns rather than rows.

The cleaned INDSTAT data are structured as:

Country | Year | ISIC4 | Employment | Output | Value Added | Wages | Establishments | ...

This ensures one observation per country–year–industry combination.

Merging Datasets

After constructing the NTM industry panel and cleaning INDSTAT and trade association data,

the datasets are merged 1-to-1 on:

- Country
- Year
- ISIC Rev. 3 (4-digit)

The result is a unified country–year–industry dataset suitable for analysis.

Theoretical Framework and Hypothesis

The project builds on political economy models of trade policy formation, including the “protection for sale” framework and related literature on lobbying and endogenous protection.

The core hypothesis is:

Industries with higher trade association density are more likely to experience greater NTM intensity.

The logic is that trade associations act as organized interest groups capable of influencing regulatory and trade policy outcomes. As tariffs decline, NTMs may serve as substitute protection instruments, particularly in industries with strong domestic organization.

A secondary hypothesis is that NTMs may cluster in “bottleneck” industries within supply chains, potentially amplifying downstream protection effects.

Initial Code

Establishing the Database Connection

The script begins by creating a connection to DuckDB. DuckDB operates as an in-process analytical database, meaning it runs directly inside Python and allows SQL queries to be executed on CSV files without loading the entire dataset into memory. This is particularly well-suited for handling large NTM datasets.

Creating the Raw Data View

The first SQL command creates a view called `ntm_raw` using the `read_csv_auto()` function.

This function:

- Automatically detects column types
- Reads the CSV file directly from disk
- Does not duplicate the file in memory

The use of `CREATE OR REPLACE VIEW` ensures that if the script is rerun, the existing view is overwritten rather than duplicated. At this stage, no transformation occurs, the dataset is simply made accessible inside DuckDB.

Cleaning and Standardizing Variables

The second SQL block creates a new view called `ntm_clean`. This view transforms and standardizes key variables:

- ReporterName is converted to uppercase and trimmed to eliminate inconsistencies in country naming.
- StartDate is cast as an integer to ensure it is treated as a numeric year variable.
- ProductCode is converted to a string and cleaned using a regular expression that removes all non-numeric characters. This ensures HS codes contain only digits, which is necessary for concordance matching.
- Nomen_Code and NTM_Code are standardized to uppercase.

The WHERE clause filters out observations with missing country or product codes, preventing invalid records from entering the aggregation stage.

Importantly, this cleaning step does not yet aggregate data, it simply prepares it for grouping later on.

Collapsing Before Crosswalk

The third SQL block creates the ntm_collapsed view by grouping the cleaned data at the level of: country, year, and hscore.

Using GROUP BY, DuckDB partitions the dataset into unique country–year–HS combinations and computes summary statistics within each group. Specifically:

- COUNT(*) measures the total number of NTM records affecting that product.
- COUNT(DISTINCT ntm_code) measures the number of unique policy measures.
- MAX(nomen_code) retains the HS nomenclature version.

This produces one observation per country–year–HS code.

Collapsing at this stage is crucial because HS codes may later map to multiple ISIC industries through the concordance. Aggregating first prevents double-counting when the crosswalk expands HS codes into industry codes.

Exporting the Output

The final COPY command writes the collapsed dataset to a CSV file. DuckDB streams the aggregated results directly to disk, producing a clean intermediate dataset (ntm_collapsed.csv) that will be used in the HS → ISIC mapping stage.

```
import duckdb

# Connect to DuckDB
con = duckdb.connect()

# -----
# Read CSV file (USE FULL PATH TO AVOID ERRORS)
# -----
con.execute(r"""
```

```

CREATE OR REPLACE VIEW ntm_raw AS
SELECT *
FROM read_csv_auto('z:/Working Data/Code/processed_ntm_wits_data.csv')
""")

# -----
# Clean + Standardize Columns
# -----

con.execute("""
CREATE OR REPLACE VIEW ntm_clean AS
SELECT
    upper(trim(ReporterName)) AS country,
    CAST(StartDate AS INTEGER) AS year,
    regexp_replace(CAST(ProductCode AS VARCHAR), '^[0-9]', '', 'g') AS
hscode,
    upper(trim(Nomen_Code)) AS nomen_code,
    upper(trim(NTM_Code)) AS ntm_code
FROM ntm_raw
WHERE ReporterName IS NOT NULL
    AND ProductCode IS NOT NULL
""")

# -----
# Collapse FIRST (Before Crosswalk)
# Final key = country + year + hscode
# -----

con.execute("""
CREATE OR REPLACE VIEW ntm_collapsed AS
SELECT
    country,
    year,
    hscode,
    MAX(nomen_code) AS nomen_code,
    COUNT(*) AS total_ntm_rows,
    COUNT(DISTINCT ntm_code) AS n_unique_ntm_codes
FROM ntm_clean
GROUP BY country, year, hscode
""")

# -----

```

```
# Save Output
# -----
con.execute("""
COPY ntm_collapsed
TO 'z:/Working Data/Code/ntm_collapsed.csv'
(HEADER, DELIMITER ',')
""")

print("Done! ntm_collapsed.csv created successfully.")
```